

## Problem 3

A body moves along a straight path, and over successive segments of equal length  $s$ , it maintains constant speeds  $v_1, v_2, v_3, \dots, v_n$ . What is the average speed of the body?

## Solution

To calculate the average speed when distances are equal, we use the formula:

$$v_{\text{avg}} = \frac{n}{\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3} + \dots + \frac{1}{v_n}},$$

where  $n$  is the number of equal segments.

## Steps

1. For two segments ( $n = 2$ ):

$$v_{\text{avg}} = \frac{2}{\frac{1}{v_1} + \frac{1}{v_2}} = \frac{2v_1v_2}{v_1 + v_2}.$$

2. For three segments ( $n = 3$ ):

$$v_{\text{avg}} = \frac{3}{\frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}}.$$

3. In the general case for  $n$  segments:

$$v_{\text{avg}} = \frac{n}{\sum_{i=1}^n \frac{1}{v_i}}.$$

## Conclusion

The average speed of the body over the entire journey depends on the harmonic mean of the segment speeds.